MIPS Delayed Trigger Capability

Delayed triggering is a capability added to the MIPS system with version 1.73 of the firmware. This capability allows you to define a trigger input and then define a delay time from the detection of the trigger until the selected module is triggered. Additionally the delayed trigger capability allows the module to be re-triggered at a user defined period for a defined number of cycles.

The delayed triggering capability must be set up using the host computer interface.

Below are the delayed trigger commands; please note the commands are case sensitive:

SDTRIGINP,*<input>*,*<level>*
This command defines the trigger *input* (Q through X) and the active *level* for the trigger, POS for positive edge or NEG for negative edge.

SDTRIGDLY,*<time>*
This command defines the trigger delay *time* in microseconds. This is the delay time from the detection of a trigger until the defined module is triggered.

GDTRIGDLY
This command returns the trigger delay time.

SDTRIGPRD,*<time>*
This command defines the repeat period *time* in microseconds. This parameter defines the time between the first trigger event and all subsequent triggers for the defined repeat count.

GDTRIGPRD
This command returns the current period time in microseconds.

SDTRIGRPT,*<count>*
This command defines the delayed trigger repeat *count*. This is the number of times a module trigger is generated for each input hardware trigger. The default for this parameter is 1, resulting in only one module trigger per input hardware trigger. The period time, discussed above, defines the time between repeated trigger events. Setting this parameter to 0 will cause the module trigger to repeat indefinitely.

GDTRIGRPT
This command returns the current repeat count.

SDTRIGMOD,*<ARB|ADC|SWEEP|AUXTRIG>*
This command defines the *module* to be triggered when a delayed trigger event occurs. Four different delayed trigger option are avalible.

ARB,	Generates an ARB sync signal to synchronize multiple ARB modules.
ADC,	Triggers the ADC system to start recording a data vector.
SWEEP,	Starts an ARB frequency/voltage sweep.
AUXTRIG,	Generate a AUXTRIG output pulse.

SDTRIGENA,<TRUE|FALSE>

This command enables or disables the delayed trigger system. The enable (TRUE) command option should only be issued after all parameters are defined and valid.

GDTRIGENA

This command returns the current status of the delayed trigger enable, TRUE or FALSE.

Table 1, Delayed Trigger Commands

Command	Parameters	Description
SDTRIGINP	<input>,<level>	Defines the trigger input (Q through X) and active level. POS = positive edge, NEG = negative edge.
SDTRIGDLY	<time>	Sets the trigger delay time in microseconds from trigger detection to module trigger.
GDTRIGDLY		Returns the current trigger delay time in microseconds.
SDTRIGPRD	<time>	Sets the repeat period time in microseconds between the first trigger event and subsequent triggers.
GDTRIGPRD		Returns the current period time in microseconds.
SDTRIGRPT	<count>	Sets the delayed trigger repeat count. Default is 1. Setting to 0 causes the module trigger to repeat indefinitely.
GDTRIGRPT		Returns the current repeat count.
SDTRIGMOD	<ARB|ADC|SWEEP|AUXTRIG>	Defines the module to trigger on a delayed trigger event.
SDTRIGENA	<TRUE|FALSE>	Enables or disables the delayed trigger system. Issue TRUE only after all parameters are defined and valid.
GDTRIGENA		Returns the current status of the delayed trigger enable: TRUE or FALSE.

Adjusting Parameters at Runtime

The delay parameters can be adjusted while the system is operating and accepting triggers. This allows you to “tune” the delay times and periods for your specific application. If you have the repeat count set to 0 (indefinite) and wish to stop, you can set the repeat count to 1 or disable the delayed trigger capability.

The delayed trigger capability is re-triggerable. This means if it is running and a new trigger is detected it will restart and synchronize with the new trigger event.

Example

Below is an example that sets up the delayed trigger to accept a trigger on the T input positive edge and trigger the ARB module 3 times. The first trigger occurs 5 milliseconds after the hardware trigger event on the T input and the next two occur at an 8-millisecond period:

```
SDTRIGINP,T,POS
SDTRIGDLY,5000
SDTRIGPRD,8000
SDTRIGRPT,3
SDTRIGMOD,ARB
SDTRIGENA,TRUE
```

This capability allows you to define a short ARB event that needs to repeat at a slow period. Note that ARB buffers can repeat and the delayed trigger also has repeat capability, so you can generate very complex patterns.

Important Notes

Modules have trigger options; for example, you can set up the ARB module for an external trigger. Make sure you do not set up the module to accept a hardware trigger using the same input used for the delayed trigger, as this will result in unpredictable outcomes. However, you can have the ARB module triggered by two separate inputs, one for the standard triggering and one for the delayed triggering.